c) Transgenic cow Rosie which produces high fat milk for making ghee
d) Animals like bulls for farm work as they have super power

2. Read the following four statements (A to D) about certain mistakes in two of them. [AIPMT MAINS 2011]
a) The first transgenic buffalo, Rosie produced milk which was human alpha – lactalbumin enriched.
b) Restriction enzymes are used in isolation of DNA from other macromolecules.
c) Downstream processing is one of the steps of rDNA technology
d) Disarmed pathogen vectors are also used in transfer of rDNA into the host.
Which of the two statements have mistakes?
a) B and C b) C and D
c) A and C **d) A and B**

3. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of [AIPMT 2013]
a) Non-recombinant bacteria containing β-galactosidase.
b) Insertional inactivation of α-galactosidase in non-recombinant bacteria.
c) Insertional inactivation of α-galactosidase in recombinant bacteria.
d) Inactivation of glycosidase enzyme in recombinant bacteria

4. Which body of the Government of India regulates GM research and safety of introducing GM organism for public services? [AIPMT 2015]
a) Bio-safety committee
b) Indian council of agricultural research
c) Genetic engineering approval committee
d) Research committee on Genetic manipulation

5. In genetic engineering, a DNS segment (gene) of interest is transferred to the host cell through a vector. Consider the following four agents (A to D) in this regard and select correct option about which one or more of these can be used as vector/vectors. [AIPMT MAIN 2010]
A) A bacterium B) Plasmid
C) Plasmodium D) Bacteriophage
a) (A), (B) and (D) only b) (A) only
c) (A) and (C) only **d) (B) and (D) only**

6. Which one of the following palindromic base sequences in DNA can be easily cut at about the middle by some particular restriction enzyme? [AIPMT PRE 2010]
a) 5' – CGTTCG – 3' 3' –ATGGTA -5'
b) 5'-GATATG -3' 3' CTACTA -5'
c) 5' –GAATTC – 3' 3' – CTTAAG-5'
d) 5' –CACGTA -3' 3' –CTCAGT -5'

7. Restriction endonucleases are enzymes which [AIPMT PRE 2010]
a) Make cuts at specific positions within the DNA molecule.
b) Recognize a specific nucleotide sequence for binding of DNA ligase.
c) Restrict the action of the enzyme DNA polymerase.
d) Remove nucleotides from the ends of the DNA molecule.

10. Stirred – tank bioreactors have been designed for [AIPMT PRE 2010]
a) Addition of preservatives of the product
b) Purification of the product
c) Ensuring anaerobic conditions in the culture vessel
d) Availability of oxygen throughout the process

11. There is a retriction endonuclease called EcoRI. What does 'co' part in it stand for? [AIPMT PRE 2011]
a) Coelom b) Coenzyme
c) Coli d) Colon

12. Which one is true state regarding DNA polymerase used in PCR? [AIPMT PRE 2012]
a) It is used to ligate introduced DNA in recipient cells.
b) It serves as selectable marker
c) It is isolated from a virus.
d) It remains active at high temperature.

13. For transformation, micro-particles coated with DNA to be bombarded with gene gun are made up of [AIPMT PRE 2012]
a) Silver or Platinum b) Platinum or Zinc
c) Silicon or Platinum **d) Gold or Tungsten**

Chapter 11 - Organisms and Population

1. Which one of the following is most appropriately defined? [AIPMT MAINS 2010]
a) Host is an organism which provides food to another organism.
b) Amensalism is a relationship in which one species is benefited whereas the other is unaffected.
c) Predator is an organism that catches and kills other organism for food.
d) Parasite is an organism which always lives inside the body of other organism and may kill it.

2. Study the four statements (1 to 4) given below and select the two correct ones out of them. [AIPMT PRE 2010]
a) A lion eating a deer and a sparrow feeding on grain are ecologically similar in being consumers.
b) Predator star fish Pisaster helps in maintaining species diversity of some invertebrates.
c) Predators ultimately lead to the extinction of prey species.
d) Production of chemicals such as nicotine, strychnine by the plants is metabolic disorders.
The two correct statements are
a) (B) and (C) b) (C) and (D)
c) (A) and (D) **d) (A) and (B)**

3. Which two of the following changes (1 to 4) usually tend to occur in the plain dwellers when they move to high altitudes(3500 m or more)? [AIPMT PRE 2010]
A) Increase in red blood cell size
B) Increase in red blood cell production
C) Increased breathing rate
D) Increase in thrombocyte count
a) (B) and (C) b) (C) and (D)
c) (A) and (D) **d) (A) and (B)**

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